Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The reduction potential (in volt) of a hydrogen electrode set up with a $2 \times 10^{-2} \mathrm{M}$ aqueous solution of a weak mono basic acid $\left(\mathrm{K}_{\mathrm{a}}=5 \times 10^{-5}\right)$ at one atmosphere and $25^{\circ} \mathrm{C}$ is

(A) $+0.09$

(B) $+0.18$

(C) -0.09

(D) -0.18

Q.2

A solution of copper sulphate $\left(\mathrm{CuSO}_4\right)$ is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at the cathode (at. mass of $\mathrm{Cu}=63 \mathrm{u}$) is :

(A) $0.3892 \mathrm{~g}$

(B) $0.2938 \mathrm{~g}$

(C) $0.2398 \mathrm{~g}$

(D) $0.3928 \mathrm{~g}$

Q.3

A battery is constructed of Cr and $\mathrm{Na}_2\mathrm{Cr}_2\mathrm{O}_7$. The unbalanced chemical equation when such a battery discharges is following:

$\mathrm{Na}_2\mathrm{Cr}_2\mathrm{O}_7+\mathrm{Cr}+\mathrm{H}^{+} \rightarrow \mathrm{Cr}^{3+}+\mathrm{H}_2\mathrm{O}+\mathrm{Na}^{+}$

If one Faraday of electricity is passed through the battery during the charging, the number of moles of Cr^{3+} removed from the solution is

(A) $\frac{4}{3}$

(B) $\frac{1}{3}$

(C) $\frac{3}{3}$

(D) $\frac{2}{3}$

Q.4

At $298 \mathrm{~K}$ the standard free energy of formation of $\mathrm{H}_2\mathrm{O}$ (l) is $-237.20 \mathrm{~kJ} / \mathrm{mol}$ while that of its ionisation into H^{+} ion and hydroxyl ions is $80 \mathrm{~kJ} /$ mol, then the emf of the following cell at $298 \mathrm{~K}$

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will be
[Take Faraday constant $\mathrm{F}=96500 \mathrm{C}$] $\mathrm{H}_2(\mathrm{~g}, 1 \mathrm{bar})\left|\mathrm{H}^{+}(1 \mathrm{M})\right|\left|\mathrm{OH}^{-}(1 \mathrm{M})\right| \mathrm{O}_2(\mathrm{~g}, 1 \mathrm{bar})$

(A) $0.40 \mathrm{~V}$

(B) $0.81 \mathrm{~V}$

(C) $1.23 \mathrm{~V}$

(D) $-0.40 \mathrm{~V}$

Q.5

The emf of a particular voltaic cell with the cell reaction $\mathrm{Hg}_2^{2+}+\mathrm{H}_2 \rightleftharpoons 2 \mathrm{Hg}+2 \mathrm{H}^{+}$ is 0.65 V. The maximum electrical work of this cell when $0.5 \mathrm{~g}$ of H_2 is consumed.

(A) $-3.12 \times 10^4 \mathrm{~J}$

(B) $-1.25 \times 10^5 \mathrm{~J}$

(C) $\quad 25.0 \times 10^6 \mathrm{~J}$

(D) None of these

Q.6

A cell reaction would be spontaneous if the cell potential and $\Delta_{\mathrm{r}} G$ are respectively :

(A) positive and negative

(B) negative, negative

(C) zero, zero

(D) positive, zero

Q.7

The Nernst equation $\mathrm{E}=\mathrm{E}^{\circ}-\mathrm{RT} / \mathrm{nF}, \ln \mathrm{Q}$ indicates that the Q will be equal to equilibrium constant K_{c} when :

(A) $\mathrm{E}=\mathrm{E}^{\circ}$

(B) $\mathrm{RT} / \mathrm{nF}=1$

(C) $\mathrm{E}=$ zero

(D) $E^{\circ}=1$

Q.8

Thermodynamic efficiency of a cell is given by:

(A) $\frac{\Delta H}{\Delta G}$

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(B) $\frac{nFE}{\Delta G}$

(C) $\frac{-nFE}{\Delta H}$

(D) nFE°

Q.9

In the electrolysis of water, one faraday of electrical energy would liberate

(A) one mole of oxygen

(B) one gram atom of oxygen

(C) 8 g oxygen

(D) 22.4 lit. of oxygen

Q.10

In the electrolysis of $CuCl_2$ solution, the mass of the cathode increased by 3.2 g. What occured at the copper anode?

(A) 0.12 litre of Cl_2 was liberated

(B) 0.56 litre of O_2 was liberated

(C) 0.1 molCu^{2+} passed into the solution.

(D) 0.05 mol of Cu^{2+} passed into the solution.

Q.11

Given $E^{\circ}_{Cu^{2+}/Cu} = 0.34$ V, $E^{\circ}_{Cu^{2+}/Cu^{+1}} = 0.15$ V
Standard electrode potential for the half cell Cu^{+}/Cu is

(A) 0.38 V

(B) 0.53 V

(C) 0.19 V

(D) 0.49 V

Q.12

At 298 K, the standard reduction potentials are 1.51 V for $MnO_4^- \mid Mn^{2+}$, 1.36 V for $Cl_2 \mid Cl^-$, 1.07 V for $Br_2 \mid Br^-$, and 0.54 V for $I_2 \mid I^-$. At $pH = 3$, permanganate is expected to oxidize $\left(\frac{RT}{F} = 0.059 \text{ V}\right)$

(A) Cl^-, Br^- and I^-

(B) Br^- and I^-

(C) Cl^- and Br^-

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(D) I^- only

Q.13

Electrode potentials (E°) are given below :

$$Cu^+ / Cu = +0.52\,V,$$
$$Fe^{3+} / Fe^{2+} = +0.77\,V,$$
$$\frac{1}{2} I_2(s) / I^- = +0.54\,V,$$
$$Ag^+ / Ag = +0.88V.$$

Based on the above potentials, strongest oxidizing agent will be:

(A) Cu^+

(B) Fe^{3+}

(C) Ag^+

(D) I_2

Q.14

Al_2O_3 is reduced by electrolysis at low potentials and high currents. If 4.0×10^4 amperes of current is passed through molten Al_2O_3 for 6 hours, what mass of aluminium is produced? (Assume 100% current efficiency. At. mass of $Al = 27\ g\ mol^{-1}$)

(A) 8.1×10^4 g

(B) 2.4×10^5 g

(C) 1.3×10^4 g

(D) 9.0×10^3 g

Q.15

Electrolysis of dilute aqueous NaCl solution was camed out by passing $10\ mA$ current. The time required to liberate $0.01\ mol$ of H_2 gas at the cathode is $(1\ F = 96500 C mol^{-1})$

(A) 9.65×10^4 s

(B) $19.3 \times 10^4 s$

(C) 28.95×10^4 s

(D) 38.6×10^4 s

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Q.16

The standard oxidation potentials of Cu/Cu^{2+} and Cu^{+}/Cu^{2+} are $-0.34\ V$ and $-0.16\ V$ respectively. The standard electrode potential of Cu^{+}/Cu would be

(A) $0.18\ V$

(B) $0.52\ V$

(C) $0.82\ V$

(D) $0.49\ V$

Q.17

Acidified water is electrolysed using an inert electrode. The volume of gases liberated at STP is $0.168\ L$. The quantity of charge passed through the acidified water would be

(A) $96,500C$

(B) $9,650C$

(C) $965C$

(D) $168C$

Q.18

The useful work done during the reaction $Ag_{(s)} + \frac{1}{2}Cl_{2(\ g)} \rightarrow AgCl_{(s)}$ Would be Given $E^{0}_{Cl_2\ |\ Cl} = 1.36\ V, E^{0}_{AgCl\ |\ Ag\ |\ Cl^{-}} = 0.220\ V$
$P_{Cl_2} = 1atm$, and $T = 298K$

(A) $110 KJmol^{-1}$

(B) $220 KJmol^{-1}$

(C) $55 KJmol^{-1}$

(D) $100 KJmol^{-1}$

Q.19

The spontaneous redox reaction/s among the following is/are
(1) $2Fe^{3+} + Fe \rightarrow 3Fe^{2+}$
(2) $Hg_2^{++} \rightarrow Hg^{++} + Hg$
(3) $3AgCl + NO + 2H_2O \rightarrow 3Ag + 3Cl^{-} + NO_3^{-} + 4H^{+}$
Given that
$E_{Fe^{+3}/Fe^{+2}} = 0.77\ V, E^{o}_{Fe^{+2}/Fe} = -0.44\ V$
$E^{o}_{Hg_2^{++}/Hg} = 0.85\ V, \quad E_{Hg^{++}/Hg_2} = 0.92\ V$
$E^{o}_{AgCl/Ag} = 0.22\ V, \quad E^{\circ}_{NO_3^{-}/NO} = 0.96\ V$

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(A) 1

(B) $1, 2, 3$

(C) 1,2

(D) 1,3

Q.20

Which one of the following metals could not be obtained on electrolysis of aqueous solution of its salts

(A) Ag

(B) Mg

(C) Cu

(D) Cr

Answers & Solutions

Q.1 Answer:

-0.18

Solution:

$$2\mathrm{H}^{+} + 2\mathrm{e}^{-} \longrightarrow \mathrm{H}_2$$

$$E_{\text{cell}} = 0.06 \log \left[\mathrm{H}^{+}\right] = 0.06 \times \log \sqrt{K_a C}$$

$$= 0.06 \times \log 10^{-3} = -0.18 \text{ V}$$

Q.2 Answer:

0.2938 g

Solution:

$\mathrm{W} = \mathrm{Zit}$

where Z = **Electrochemical equivalent Eq. wt. of copper** $= \frac{63}{2} = 31.5$

$Z = \frac{31.5}{96500}$

$\mathrm{W} = \mathrm{Zit} = \frac{31.5}{96500} \times 1.5 \times 10 \times 60 = 0.2938$ g

Q.3 Answer:

$\frac{3}{3}$

Solution:

Reduction half reaction :

$\mathrm{Cr_2O_7^{2-}} + 6e^{-} + 14\mathrm{H}^{+} \longrightarrow 2\mathrm{Cr}^{3+} + 7\mathrm{H_2O}$

Oxidation half reaction :

$\mathrm{Cr} \longrightarrow \mathrm{Cr}^{3+} + 3e^{-}$

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Overall reaction :

$\mathrm{Cr_2O_7^{2-} + Cr + 14H^+ + 3}e^- \longrightarrow \mathrm{3Cr^{3+} + 7H_2O}$

3 F of electricity = 3 moles of $\mathrm{Cr^{3+}}$ 1F of electricity = $\frac{3}{3}$ moles of $\mathrm{Cr^{3+}}$

Q.4 Answer:

$0.40\ \mathrm{V}$

Solution:

Cell reaction cathode :

$\mathrm{H_2O(l) + \frac{1}{2}O_2(\ g) + 2e^- \longrightarrow 2OH^-(aq)}$

anode: $\mathrm{H_2(\ g) \longrightarrow 2H^+(aq) + 2e^-}$

$\mathrm{H_2O(l) + \frac{1}{2}O_2(\ g) + H_2(\ g) \longrightarrow 2H^+(aq) + 2OH^-(aq)}$

Also we have $\mathrm{H_2(\ g) + \frac{1}{2}O_2(\ g) \longrightarrow H_2O(l)}$

$\Delta \mathrm{G_f^\circ = -237.2\ kJ/mole}$

$\mathrm{H_2O(l) \longrightarrow H^+(aq) + OH^-(aq)}$

$\Delta \mathrm{G^\circ = 80\ kJ/mol}$

Hence for cell reaction

$\Delta \mathrm{G^\circ = -237.2 + (2 \times 80) = -77.20\ kJ/mol}$

$\therefore \quad \mathrm{E^\circ = -\frac{\Delta G^\circ}{nF} = \frac{77200}{2 \times 96500} = 0.40\ V}$

Q.5 Answer:

$-3.12 \times 10^4\ \mathrm{J}$

Solution:

$W_{\max} = -n \cdot FE$

$W_{\max} = -2 \times 96500 \times 0.65 = -1.25 \times 10^5\ \mathrm{J}$

$0.5\ \mathrm{gH_2} = 0.25\ \mathrm{mole}$

Hence $W_{\max}$

$= -1.25 \times 10^5 \times 0.25 = -3.12 \times 10^4\ \mathrm{J}$

Q.6 Answer:

positive and negative

Solution:

For spontaneous $\mathrm{\Delta_r G^\circ = -nFE^\circ < 0}$

$\therefore \mathrm{E^\circ > 0}$

Q.7 Answer:

$\mathrm{E = zero}$

Solution:

When $\mathrm{Q = K_C, E = 0}$

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Q.8 Answer:

$\frac{-\mathrm{nFE}}{\Delta \mathrm{H}}$

Solution:

Thermodynamic efficiency of a cell is given by the ratio of Gibbs function change to enthalpy change in the overall cell reaction.

Q.9 Answer:

$8\ \mathrm{g}$ oxygen

Solution:

According to the definition $1\ \mathrm{F}$ **or** $96500 \mathrm{C} 1\ \mathrm{s}$ **the charge carried by** $1\ \mathrm{mol}$ **of electrons when water is electrolysed**

$2 \mathrm{H}_2 \mathrm{O} \longrightarrow 4 \mathrm{H}^{+}+\mathrm{O}_2+4 \mathrm{e}^{-}$

So, 4 Faraday of electricity liberate $=32\ \mathrm{g}$ **of** O_2 **Thus 1 Faraday of electricity liberate**

$=\frac{32}{4}\ \mathrm{g}$ **of** $\mathrm{O}_2=8\ \mathrm{g}$ **of** O_2

Q.10 Answer:

$0.05\ \mathrm{mol}$ of Cu^{2+} passed into the solution.

Solution:

The amount. of copper deposited at cathode byreduction of Cu^{2+} **ions is** $\frac{3.2}{63}=0.05$ **moles.**

The same amount 0.05 **mole of** Cu^{2+} **must pass into solution from anode by oxidation**

Q.11 Answer:

$0.53\ \mathrm{V}$

Solution:

$\mathrm{Cu}^{2+}+e^{-} \longrightarrow \mathrm{Cu}^{+}$

$E_1^0=0.15\ \mathrm{V} ; \Delta G_1^0=-n_1 E_1^0 F$

$\mathrm{Cu}^{2+}+2e \longrightarrow \mathrm{Cu}$

$E_2^{\mathrm{o}}=0.34\ \mathrm{V} ; \Delta G_2^0=-n_2 E_2^0 F$

On subracting eq.(i) from eq. (ii) we get

$\mathrm{Cu}^{+}+e^{-} \longrightarrow \mathrm{Cu}; \quad \Delta G^0=\Delta G_2^{\mathrm{O}}-\Delta G_1^{\mathrm{o}}$

$-n E^0 F=-\left(n_2 E^0 F-n_1 E_1^0 F\right)$

$E^{\circ}=\frac{n_2 E_2^0 F-n_1 E_1^0 F}{n F}$

$=\frac{2 \times 0.34-0.15}{1}=0.53\ \mathrm{V}$

Q.12 Answer:

Br^{-} and I^{-}

Solution:

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$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

$E = 1.51 - \frac{0.059}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8}$

Taking Mn^{2+} and MnO_4^- in standard state i.e. 1M,

$E = 1.51 - \frac{0.059}{5} \times 8 \log \frac{1}{[H^+]}$

$= 1.51 - \frac{0.059}{5} \times 8 \times 3 = 1.2268$ V

Hence at this pH, MnO_4^- will oxidise only Br^- and I^- as SRP of Cl_2/Cl^- is 1.36 V which is greater than that for MnO_4^-/Mn^{2+}.

Q.13 Answer:

Ag^+

Solution:

Higher the value of reduction potential stronger will be the oxidising hence based on the given values Ag^+ will be strongest oxidizing agent.

Q.14 Answer:

8.1×10^4 g

Solution:

$\because Q = i \times t$

$\therefore Q = 4.0 \times 10^4 \times 6 \times 60 \times 60C = 8.64 \times 10^8 C$

Now since 96500 C liberates 9 g of Al 8.64×10^8C liberates

$\frac{9}{96500} \times 8.64 \times 10^8$ g Al $= 8.1 \times 10^4$ g **of A**

Q.15 Answer:

$19.3 \times 10^4 s$

Solution:

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

1 molH_2 **requires 2 faraday current**

$\therefore 0.01$ mol **will require** $= 0.02$ **faraday current.**

$Q = 0.02 \times 96500C$

$Q = i \times t$

$i = 10$ **mili ampere** $= 10 \times 10^{-3}$ **ampere**

$Q = 10 \times 10^{-3} \times t$

$\Rightarrow 0.02 \times 96500 = 10^{-2} \times t$

$\Rightarrow t = 19.30 \times 10^4$sec

Option B is correct.

Q.16 Answer:

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$0.52\ \mathrm{V}$

Solution:

Reactions

$\mathrm{Cu}^{2+} + 2\mathrm{e}^- \rightarrow \mathrm{Cu} \quad \Delta \mathrm{G}^0 = -\mathrm{nFE}^0$

$\mathrm{Cu}^+ \rightarrow \mathrm{Cu}^{2+} + \mathrm{e}^-$

$\Delta \mathrm{G}_{\mathrm{I}}^0 = -2\ \mathrm{F} \times 0.34 = 0.68\ \mathrm{F}$

$\Delta \mathrm{G}_{\mathrm{II}}^0 = \mathrm{F} \times 0.16 = 0.16\ \mathrm{F}$

Adding, we get

$\mathrm{Cu}^+ + \mathrm{e}^- \rightarrow \mathrm{Cu} \Delta \mathrm{G}_{\mathrm{III}}^0 = \Delta \mathrm{G}_{\mathrm{I}}^0 + \Delta \mathrm{G}_{\mathrm{I}}^0$

$= 0.52\ \mathrm{F}$

$= -\mathrm{FE}^0$

$\mathbf{F}^0 = 052\ \mathrm{V}$

Q.17 Answer:

$965C$

Solution:

$\underset{\mathbf{2x}}{\mathbf{2H_2O}} \rightarrow \underset{\mathbf{x}}{\mathbf{2H}_{2(\mathrm{g})}} + \mathrm{O}_{2(\mathrm{g})}$

$\therefore 3x = 0.168$

$\therefore x = 0.056\ \mathrm{L}$

$\mathrm{V}_{\mathrm{H}_2} = 2\mathrm{x} = 0.112\ \mathrm{L},\ \mathrm{V}_{\mathrm{O}_2} = \mathrm{x} = 0.056\ \mathrm{L}$

$11.2\ \mathrm{L}$ **of** H_2 **at** $\mathrm{STP} \equiv \mathbf{1F}$

$0.112\ \mathrm{L}$ **of** H_2 **at** $\mathrm{STP} \equiv 0.01\ \mathrm{F}$

$0.056\ \mathrm{L}$ **of** O_2 **at** $\mathrm{STP} = 0.01\ \mathrm{F}$

$\therefore$ **The amount of electricity passed**

$= 0.01\ \mathrm{F} = 965\mathrm{C}$

Q.18 Answer:

$110\mathrm{KJmol}^{-1}$

Solution:

For the cell reaction

$\mathrm{Ag}_{(\mathrm{s})} + \frac{1}{2}\mathrm{Cl}_{2(\mathrm{g})} \rightarrow \mathrm{AgCl}_{(\mathrm{s})}$

$\mathrm{E}^0 = -1.14\ \mathrm{V}$

or $\mathrm{E} = \mathrm{E}^0 - \frac{0.0592}{1} \log \mathrm{P}_{\mathrm{Cl}_2}^{1/2}$

Under standard conditions, $\mathrm{P}_{\mathrm{Cl}_2} = 0$

$\therefore \log \mathrm{P}_{\mathrm{Cl}_2}^{1/2} = 0$

$\therefore$ **Useful work** $= -\mathbf{W}_{\max} = -\mathrm{nFE}$

$= (-1) \times (-1.14) \times 96500 \times 10^{-3}\mathrm{KJ}$

$= 110\mathrm{KJmol}^{-1}$

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Q.19 Answer:

1

Solution:

1) $2Fe^{+3} + Fe \rightarrow 2Fe^{++} + Fe^{+}$

$E^0_{rration} = E^0_{OA} - E^0_{R.A}$

$= 0.77 - (-0.44)$

$= +1.21\ V$

E^0 **is positive so** ΔG **is negative**

$\therefore$ **Spontaneous reaction**

2) $Hg_2{}^{++} \rightarrow Hg^{++} + Hg$

$E^0_{\text{radax Reaction}} = E^0_{O.A} - E^0_{RA}$

$= 0.85 - 0.92$

$= -0.07$

E^0 **is negative so** ΔG **is positive**

$\therefore$ **non-spontaneous reaction**

3) $3AgCl + NO + 2H_2O \rightarrow 3Ag + 3Cl^- + NO_3^- + 4H^+$

$E^0_{\text{redax Reertion}} = E^0_{0.A} - E^0_{RA}$

$= 0.22 - 0.96 = -074.V$

E^0 **is negative so** ΔG **is positive**

$\therefore$ **non-spontaneous reaction**

Q.20 Answer:

Mg

Solution:

The reduction potential of Mg **is less than that of water** $(E^\circ = -0.83\ V)$**. Hence their ions in the aqueous solution cannot be reduced instead water will be reduced**

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

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